

JEE MAIN REVISION MASTER: PROPERTIES OF TRIANGLES 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Properties Of Triangles - Core Application	4 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Properties Of Triangles - Core Application (Moderate Yield — 4 PYQs)

Core Principle: Key problem pattern in Kcet Properties Of Triangles.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is..." [Ans: A]
- [2024 Shift] Q#2 (MCQ): "If $\triangle ABC$ is right angled at C , then the value of $\tan A + \tan B$ is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The area of a triangle with vertices $(-3,0)$, $(3,0)$ and $(0,k)$ is 9 sq units, find the value of k is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The angles of a triangle are in A.P and the greatest angle is double the least angle, then sine of the third angle is

- (A) $\frac{\sqrt{3}}{2}$ [Correct]
- (B) $\frac{1}{2}$
- (C) $\frac{1}{\sqrt{2}}$
- (D) 0

Correct Answer: A

Solution Analysis: Let the three angles of the triangle be in A.P. So we can take them as $a-d, a, a+d$. Since the sum of angles of a triangle is 180° , $(a-d) + a + (a+d) = 180^\circ$ $3a = 180^\circ$ $a = 60^\circ$. So the middle angle is 60° . Now, the greatest angle is double the least angle. Greatest angle $= a+d$ Least angle $= a-d$. So, $a+d = 2(a-d)$. Putting $a = 60^\circ$, $60^\circ + d = 2(60^\circ - d)$ $60^\circ + d = 120^\circ - 2d$ $3d = 60^\circ$ $d = 20^\circ$. Thus the angles are $40^\circ, 60^\circ, 80^\circ$. The third angle in the A.P. is the middle angle, 60° . So, $\sin 60^\circ = \frac{\sqrt{3}}{2}$.